CLASS 8 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

# Chapter 20: Area of a Trapezium and a Polygon

Topic-wise short notes with solved examples

Areas of triangles, parallelograms, rhombuses, trapeziums and general polygons, the field-book method for irregular fields, and the circumference and area of a circle.

## 1 Triangles and rectangles

| Figure | Perimeter | Area |
|---|---|---|
| Rectangle, $l$ by $b$ | $2(l+b)$ | $l \times b$ |
| Square, side $a$ | $4a$ | $a^2$, also $\frac{1}{2} \times \text{diagonal}^2$ |
| Triangle, base $b$ and height $h$ | Sum of the sides | $\frac{1}{2} \times b \times h$ |
| Triangle with sides $a, b, c$ | $a+b+c$ | $\sqrt{s(s-a)(s-b)(s-c)}$, $s = \frac{a+b+c}{2}$ |
| Equilateral triangle, side $a$ | $3a$ | $\frac{\sqrt{3}}{4}a^2$ |

**Example 1.1** Find the area of a triangle with sides 10 cm, 17 cm and 21 cm.

SOLUTION

$s = 24$. Area $= \sqrt{24 \times 14 \times 7 \times 3} = \sqrt{7056}$.

**84 sq cm**

**Example 1.2** The base of a triangle is 16 cm and its area is 96 sq cm. Find its height.

SOLUTION

$\frac{1}{2} \times 16 \times h = 96$

**12 cm**

## 2 Parallelogram and rhombus

- **Parallelogram**: area = base × height, where the height is the perpendicular distance between the base and the opposite side.
- **Rhombus**: area = base × height, and also $\frac{1}{2} \times d_1 \times d_2$.

**Example 2.1** A parallelogram has sides 20 cm and 12 cm. The height on the 20 cm side is 9 cm. Find the area and the height on the 12 cm side.

SOLUTION

Area $= 20 \times 9 = 180$ sq cm. Then $12 \times h = 180$.

**180 sq cm; 15 cm**

**Example 2.2** The diagonals of a rhombus are 30 cm and 16 cm. Find its area, side and height.

SOLUTION

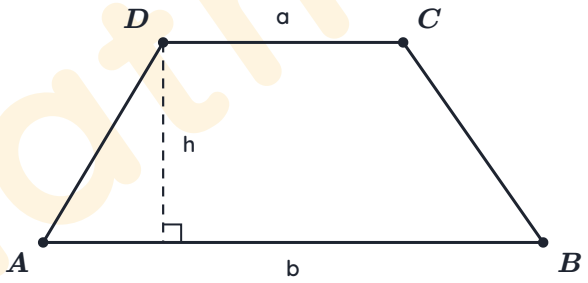
Area $= \frac{1}{2} \times 30 \times 16 = 240$ sq cm. Side $= \sqrt{15^2 + 8^2} = 17$ cm. Height $= 240 \div 17$.

**240 sq cm; 17 cm; about 14.1 cm**

## 3 Trapezium

- A trapezium has one pair of parallel sides. If their lengths are $a$ and $b$ and the perpendicular distance between them is $h$:

$$\text{Area} = \frac{1}{2} \times (a + b) \times h$$




Area of a trapezium = ½ (a + b) × h

- Reason: a diagonal splits the trapezium into two triangles with bases $a$ and $b$ and the same height $h$.

**Example 3.1** The parallel sides of a trapezium are 18 cm and 12 cm and the distance between them is 8 cm. Find its area.

SOLUTION

$\frac{1}{2} \times 30 \times 8$

**120 sq cm**

**Example 3.2** The area of a trapezium is 180 sq cm and its height is 9 cm. One parallel side is 6 cm longer than the other. Find the parallel sides.

SOLUTION

$\frac{1}{2}(x + x + 6) \times 9 = 180$, so $2x + 6 = 40$

**17 cm and 23 cm**

**Example 3.3** The parallel sides of an isosceles trapezium are 20 cm and 10 cm and each of the other sides is 13 cm. Find its area.

SOLUTION

Each end triangle has base $\frac{20 - 10}{2} = 5$ cm and hypotenuse 13 cm, so the height is 12 cm.

$\frac{1}{2} \times 30 \times 12$

**180 sq cm**

## 4 Quadrilateral

- If a diagonal has length $d$ and the perpendiculars on it from the other two vertices are $h_1$ and $h_2$:

$$\text{Area} = \frac{1}{2} \times d \times (h_1 + h_2)$$

**Example 4.1** In quadrilateral ABCD, diagonal $AC = 24$ cm and the perpendiculars from B and D on AC are 8 cm and 13 cm. Find the area.

SOLUTION

$\frac{1}{2} \times 24 \times 21$

**252 sq cm**

## 5 Polygon and field book

- The area of any polygon is found by dividing it into triangles, rectangles and trapeziums and adding their areas.
- **Regular hexagon** of side $a$: six equilateral triangles, area $= 6 \times \frac{\sqrt{3}}{4}a^2 = \frac{3\sqrt{3}}{2}a^2$.
- **Field book**: a surveyor measures along a base line and records the perpendicular offsets to the corners on each side. The field is then a set of right triangles and trapeziums.

**Example 5.1** In a field book, base line AD $= 100$ m. At 30 m from A there is an offset of 20 m to B on the left; at 60 m there is an offset of 30 m to E on the right; at 80 m there is an offset of 40 m to C on the left. Find the area of the field ABCDE.

SOLUTION

Left side: triangle (base 30, height 20) $= 300$; trapezium between 30 m and 80 m with parallel sides 20 and 40 $= \frac{1}{2} \times 60 \times 50 = 1500$; triangle (base 20, height 40) $= 400$.

Right side: triangle AED with base 100 and height 30 $= 1500$.

$300 + 1500 + 400 + 1500 = 3700$ **sq m**

**Example 5.2** Find the area of a regular hexagon of side 6 cm.

SOLUTION

$\frac{3\sqrt{3}}{2} \times 36$

$54\sqrt{3}$ **sq cm, about 93.5 sq cm**

## 6 Circle

- Circumference $= 2\pi r = \pi d$. Area $= \pi r^2$. Take $\pi = \frac{22}{7}$ unless told otherwise.
- **Ring** between two concentric circles of radii $R$ and $r$: area $= \pi(R^2 - r^2)$.
- A wheel covers a distance equal to its circumference in one revolution.

**Example 6.1** Find the circumference and the area of a circle of radius 14 cm.

SOLUTION

$2 \times \frac{22}{7} \times 14$ and $\frac{22}{7} \times 196$

**88 cm and 616 sq cm**

**Example 6.2** The area of a circle is 154 sq cm. Find its radius and circumference.

SOLUTION

$r^2 = 154 \times \frac{7}{22} = 49$

**7 cm; 44 cm**

**Example 6.3** A circular path 3.5 m wide runs round a circular garden of radius 10.5 m. Find the area of the path.

SOLUTION

$\frac{22}{7}(14^2 - 10.5^2) = \frac{22}{7} \times 85.75$

**269.5 sq m**

**Example 6.4** The wheel of a cart has diameter 1.4 m. How many revolutions does it make in travelling 1.1 km?

SOLUTION

Circumference $= 4.4$ m. $1100 \div 4.4$

**250**

## 7 Paths

- Path of width $w$ outside a rectangle $l$ by $b$: area $= (l + 2w)(b + 2w) - lb$.
- Path of width $w$ inside: area $= lb - (l - 2w)(b - 2w)$.

**Example 7.1** A room 12 m by 9 m has a carpet that leaves a border 0.5 m wide all round. Find the area of the border.

SOLUTION

Carpet: 11 m by 8 m $= 88$ sq m. Room: 108 sq m.

**20 sq m**

## 8 Common mistakes

- Using a slant side as the height.
- Forgetting the $\frac{1}{2}$ in the triangle, rhombus and trapezium formulas.
- Using the diameter instead of the radius in $\pi r^2$.
- Writing the area of a ring as $\pi(R - r)^2$.
- In field-book problems, using distances from A as the heights of the trapeziums instead of the differences between them.
- Mixing units such as centimetres and metres.